

ROYAL UNIVERSITY OF PHNOM PENH

Department of Faculty of Engineering

A Comparative Study of Two-Phase Transfer Learning And K-Fold Ensemble EfficientNet-B3 with Test Time Augmented(TTA) And Threshold Optimization for Automation Diabetic Retinopathy from Retinal Fundus Images

Uon Chamroeun

Advisor: Dr Srun Sovila

Master of Information Technology Engineering

March 2026



Presentation Outline

1

Background & Motivation

Global DR challenge and AI opportunity

2

Problem Statement & Objectives

Research gaps and goals

3

Methodology

Architecture, training, and threshold optimization

4

Experiments & Results

Binary classification and ordinal grading performance

5

Discussion

Comparison, ablation, and clinical implications

6

Conclusion & Future Work

Key contributions and next steps



Background & Motivation

537M

Adults with diabetes globally
(IDF 2021)

103M

People affected by
diabetic retinopathy

65-85%

Inter-observer agreement
among human graders

783M

Projected diabetes cases
by 2045

AI-powered screening is critical to bridge the ophthalmologist shortage in low/middle-income countries

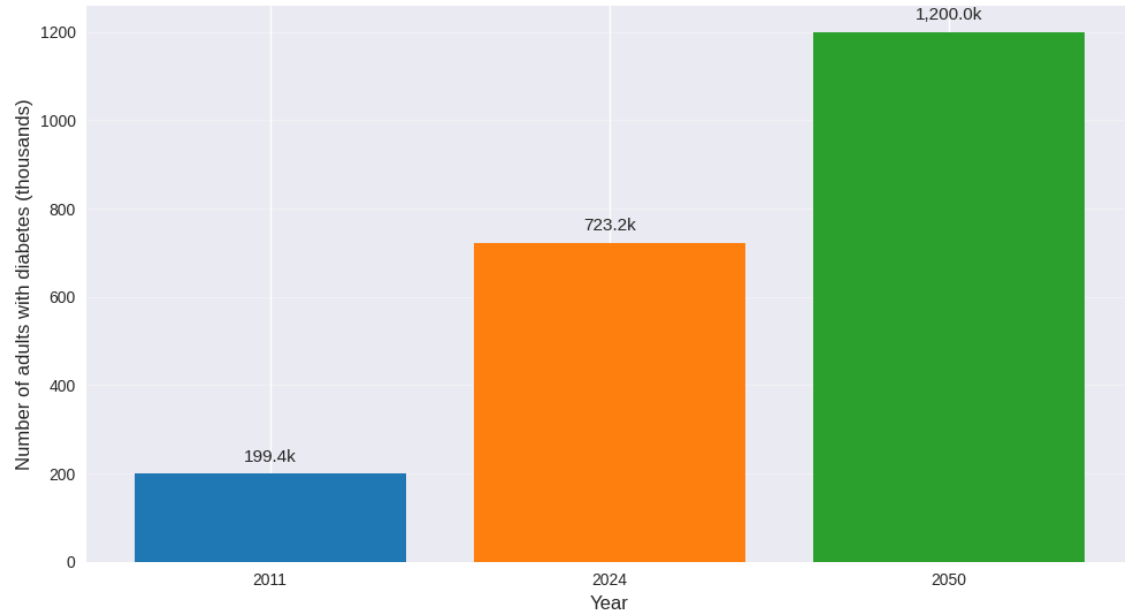


Background & Motivation

Cambodia



Number of adults (20–79 years) with diabetes in Cambodia



AI-powered screening is critical to bridge the ophthalmologist shortage in low/middle-income countries



Problem Statement

Gap 1: No Head-to-Head Comparison

No controlled comparison of two-phase transfer learning vs. K-fold ensemble on the same architecture and dataset

Gap 2: Threshold Optimization with Test Time Augmented (TTA)

For enhance performance detection of DR

Gap 3: Efficiency vs. Robustness

Single-model efficiency versus ensemble robustness tradeoff



Research Objectives

1

Compare two-phase TL vs. 5-fold ensemble using the same EfficientNet-B3 backbone and dataset

2

Develop threshold framework: binary screening and probability of grade DR

3

Quantify the synergy of TTA + threshold optimization through ablation study

4

Comprehensive evaluation: AUC-ROC, Accuracy, Cohen's Kappa, bootstrap CI 95%...



Dataset: APTOS 2019

3,534 retinal fundus images

from rural India clinics

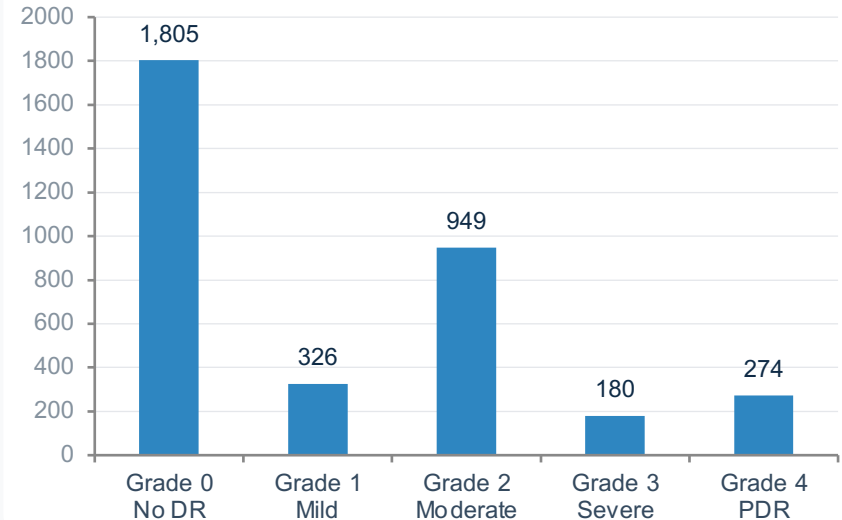
5-grade DR severity scale (Grade 0-4)

Split: 70% train (2,473) / 15% val (530) / 15% test (531)

Binary test: 270 No DR vs 261 DR Present

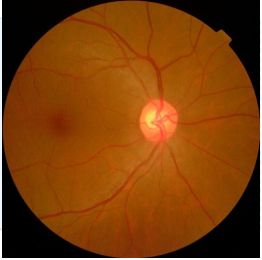
Grade	%
Grade 0 (No DR)	50.9%
Grade 1 (Mild)	9.2%
Grade 2 (Moderate)	26.9%
Grade 3 (Severe)	5.1%
Grade 4 (PDR)	7.9%

Class Distribution





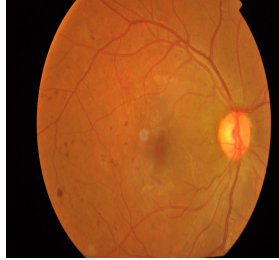
DR Severity Grades



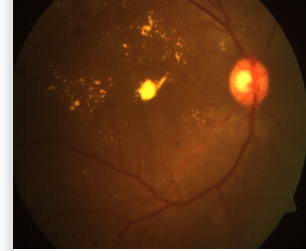
Grade 0
Normal



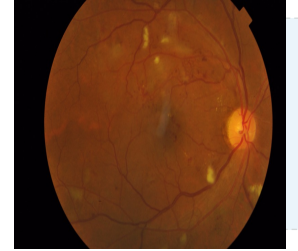
Grade 1
Mild NPDR



Grade 2
Moderate NPDR



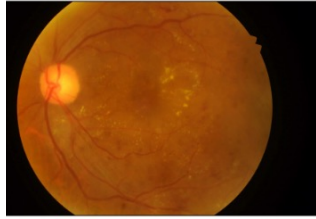
Grade 3
Severe NPDR



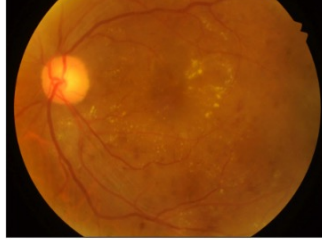
Grade 4
PDR



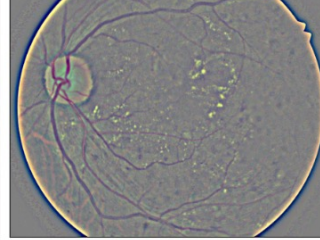
Preprocessed Dataset pipeline



4288×2848



3571×2848



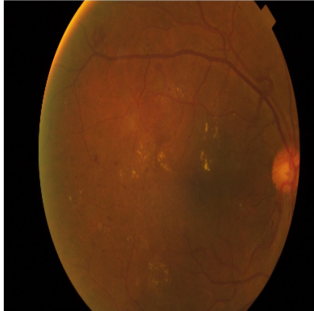
3571×2848



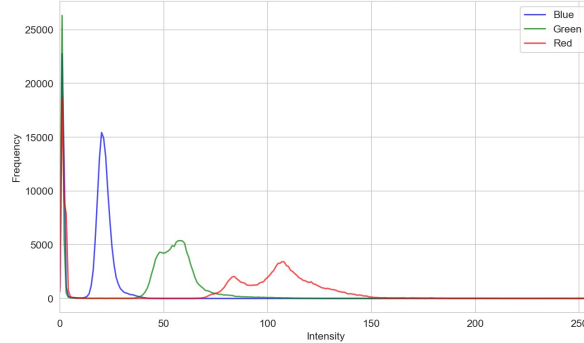
3571×2848



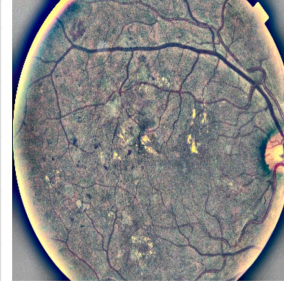
Original Image



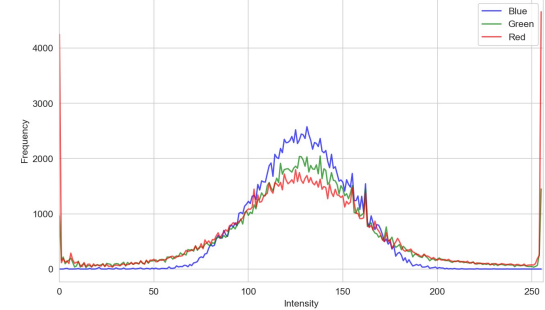
Original — Pixel Intensity



Preprocessed Image



Preprocessed — Pixel Intensity





Base Architecture: EfficientNet-B3

384 x 384

pixels

Input Resolution

1,536

features

Backbone Output

~12M

total

Parameters

ImageNet

-1K

timm library

Pretrained

Compound Scaling Strategy

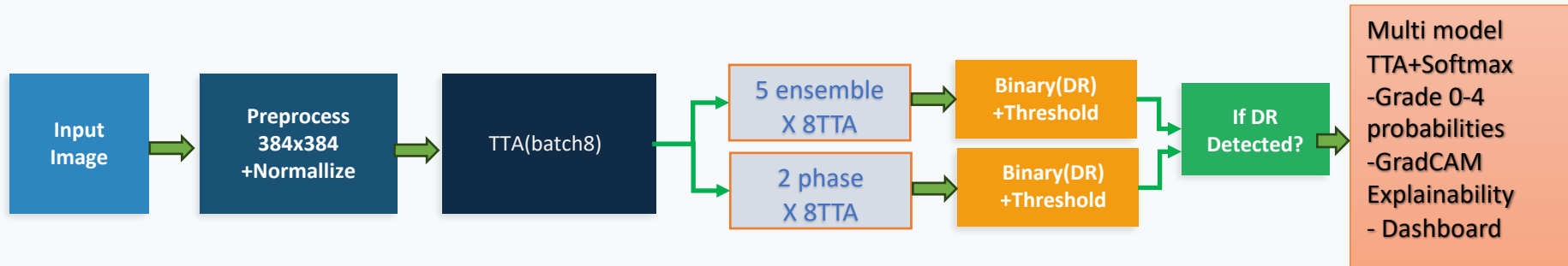
EfficientNet uses compound scaling to jointly optimize three dimensions:

Depth (more layers) x **Width** (more channels) x **Resolution** (higher input size)

B3 chosen for best accuracy-efficiency tradeoff: strong performance without excessive compute cost



Overall Pipeline



Two Parallel Strategies:

Strategy A: Two-Phase Transfer Learning

Phase 1: Freeze backbone, train head only

Phase 2: Unfreeze all, differential LR

$$P_{TTA}(x) = \frac{1}{M} \sum_{m=1}^M \sigma(f(T_m(x)))$$

Single model | 1 forward pass
T=0.1085

Strategy B: 5-Fold Ensemble

5 StratifiedKFold models

$$P_{combined}(x) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{M} \sum_{m=1}^M \sigma(f_k(T_m(x))) \right]$$

5 forward passes | Prediction variance
T=0.3597



Model 1: Two-Phase Transfer Learning

Phase 1: Head-Only Training

Backbone: **FROZEN**

Epochs: 5

Learning Rate: 1e-3

Optimizer: Adam

Head: Dropout(0.4) → Linear(1536, 1)

Only 1,537 trainable parameters

Phase 2: Full Fine-Tuning

All layers: **UNFROZEN**

Differential LR:

Backbone: 1e-5 | Head: 1e-4

Scheduler: CosineAnnealing

WarmRestarts (T0=10, Tmult=2)

Early stopping: patience=10 on val AUC

Key Advantage

Two-phase approach prevents catastrophic forgetting by stabilizing the classification head before fine-tuning the backbone. This enables efficient knowledge transfer from ImageNet to the medical imaging domain with minimal risk of divergence.



Model 2: 5-Fold Ensemble

Training Configuration

Splitting: StratifiedKFold on dr_grade (5 splits)

Epochs: 30 per fold

Optimizer: AdamW

Scheduler: OneCycleLR (max_lr=1e-3, pct_start=0.1)

Gradient Clipping: max_norm=1.0

Sampling: WeightedRandomSampler + BCEWithLogitsLoss

Deeper Classification Head

1536 features
LayerNorm
256 → GELU
Dropout
64 → GELU
Dropout
1 output

Ensemble Inference:

$$\hat{P} = \frac{1}{5} \sum_{k=1}^{k=5} \sigma(f_k(x)). \quad \text{Soft voting over 5 fold-specific models}$$



Data Augmentation

Training Augmentations (Albumentations)

HorizontalFlip	p=0.5
VerticalFlip	p=0.5
RandomRotate90	p=0.5
ShiftScaleRotate	shift=0.1, scale=0.15, rot=45°, p=0.5
RandomBrightnessContrast	± 0.2 , p=0.3
CoarseDropout	8 holes, 32x32, p=0.3

Normalization

ImageNet mean/std

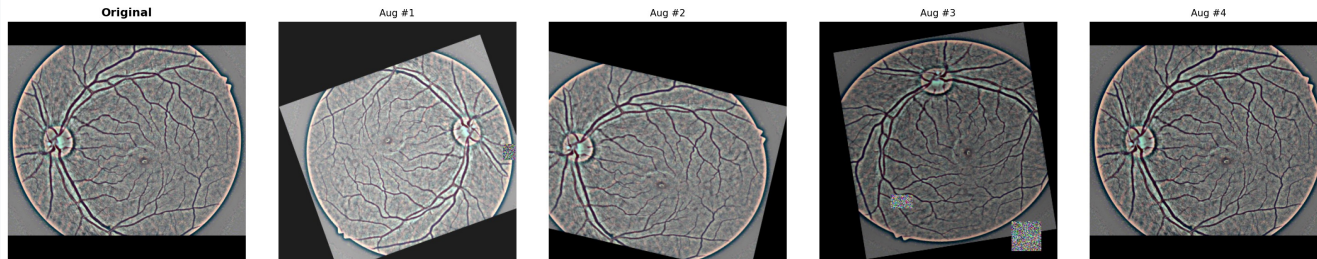
mean = [0.485, 0.456, 0.406]

std = [0.229, 0.224, 0.225]

Val/Test Transforms

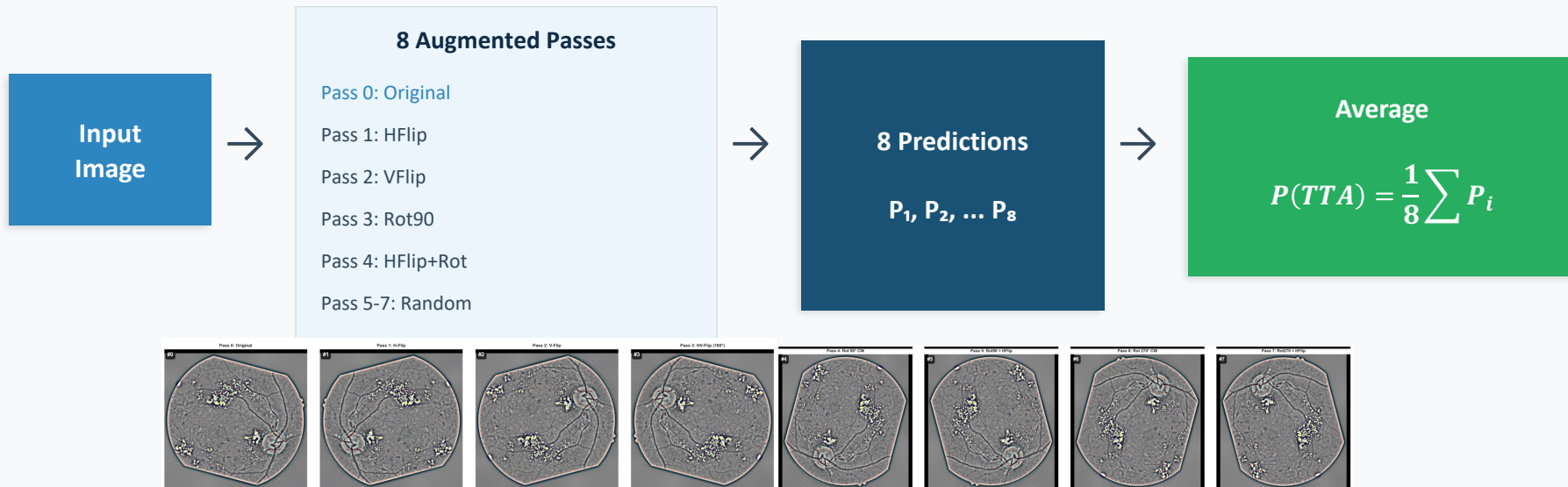
Resize(384) + Normalize only

No random augmentations





Test-Time Augmentation (TTA)



Ensemble + TTA: 5 models x 8 passes = 40 forward passes per image

TTA provides a free accuracy boost by averaging out augmentation-induced prediction variance. Combined with ensemble averaging, it maximizes robustness at the cost of inference time.



Threshold Optimization

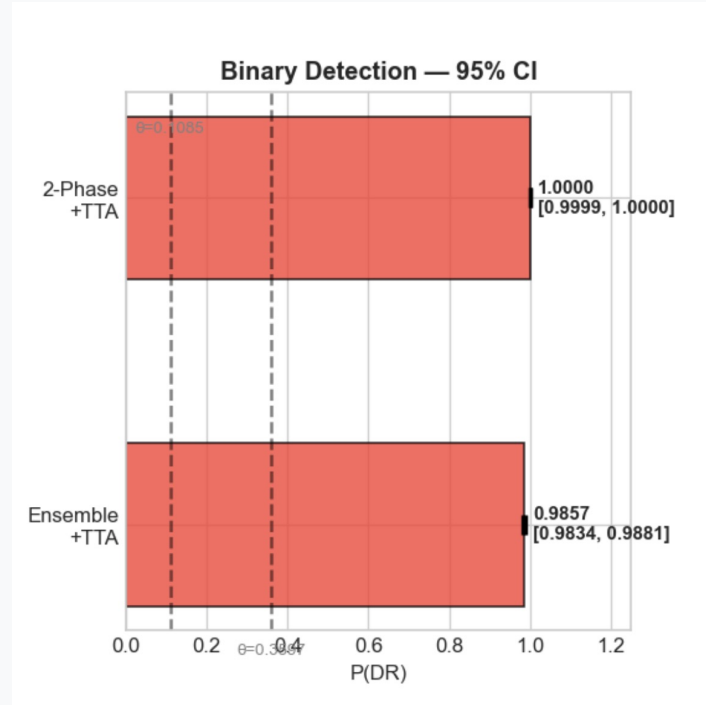
Binary Threshold (Youden's J)

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

$$t(2 \text{ phase} + \text{TTA}) = 0.1085$$

$$T(\text{Ensemble} + \text{TTA}) = 0.3584$$

Optimizes the cutpoint to maximize the sum of sensitivity and specificity





Main Results: Binary Classification

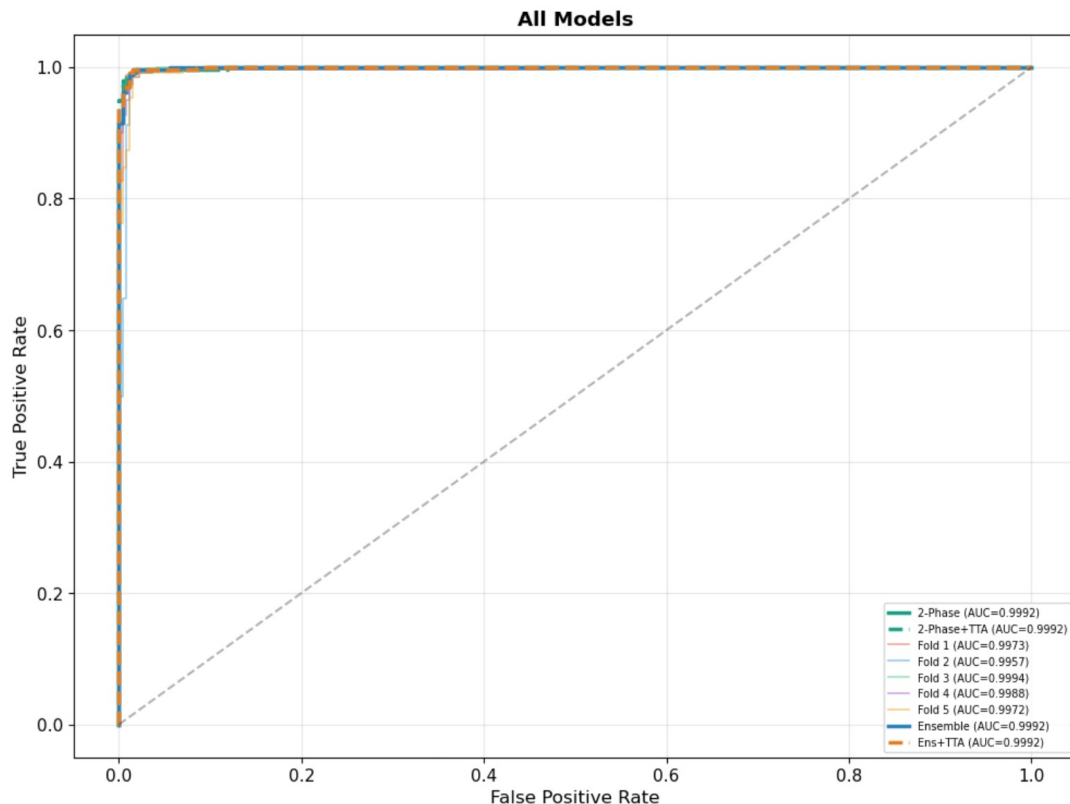
KEY SLIDE

Configuration	AUC-ROC	Accuracy	Sensitivity	Specificity	F1	Kappa
2-Phase (no TTA)	0.9992	0.9906	0.9923	0.9889	0.9904	0.9812
2-Phase + TTA	0.9993	0.9925	0.9939	0.9911	0.9923	0.9849
2-Phase + TTA + OptThr	0.9993	0.9943	0.9962	0.9926	0.9942	0.9887
Fold 1	0.9977	0.9887	0.9885	0.9889	0.9885	0.9774
Fold 2	0.9971	0.9849	0.9847	0.9852	0.9846	0.9698
Fold 3	0.9994	0.9849	0.9847	0.9852	0.9846	0.9698
Ensemble (no TTA)	0.9992	0.9887	0.9885	0.9889	0.9885	0.9774
Ensemble + TTA	0.9993	0.9906	0.9923	0.9889	0.9904	0.9812
Ensemble + TTA + OptThr	0.9994	0.9925	0.9962	0.9889	0.9923	0.9849

Both best configs exceed 99.2% accuracy and 99.6% sensitivity — highlighted rows show optimal configurations



ROC Curve Analysis



AUC-ROC Summary

2-Phase

0.9992

2-Phase+TTA

0.9993

Ensemble

0.9992

Ensemble+TTA

0.9993

Best Fold (3)

0.9994



Ablation Study

Two-Phase Path

Base (no TTA)	0.9906 acc
↑ + TTA	+0.0019 acc
↑ + OptThr	+0.0018 acc
Combined	0.9943 acc

Ensemble Path

Base (no TTA)	0.9887 acc
↑ + TTA	+0.0019 acc
↑ + OptThr	+0.0019 acc
Combined	0.9925 acc

Both TTA and threshold optimization provide additive, independent improvements

Combined effect: +0.0037 accuracy, +0.0039 sensitivity improvement over base models



Comparison with Prior Work

Study	Architecture	Dataset	AUC-ROC	Sensitivity	Specificity
Gulshan et al. (2016)	Inception-v3	EyePACS	0.991	0.974	0.934
Abràmoff et al. (2018)	Custom CNN	IDx-DR Trial	—	0.872	0.907
Ting et al. (2017)	VGGNet-19	SiDRP	0.936	0.905	0.916
APTOS 2019 Winner	Ensemble (5+)	APTOS 2019	~0.999	—	—
Ours: 2-Phase+TTA+Opt	EfficientNet-B3	APTOS 2019	0.9993	0.9962	0.9926
Ours: Ens+TTA+Opt	EffNet-B3 ×5	APTOS 2019	0.9994	0.9962	0.9889

Important Caveat

Direct comparison across studies is limited due to different datasets, population demographics, and evaluation protocols. Our results on APTOS 2019 demonstrate strong performance within this benchmark, but external validation on diverse datasets is needed to confirm generalizability.



Clinical Implications



NHS DESP Requirement

Required: >80% sensitivity

99.6%

EXCEEDS



AAO Recommendation

Required: >90% sensitivity

99.6%

EXCEEDS



IDx-DR (FDA-approved)

Required: 87.2% sensitivity

99.6%

EXCEEDS

Specificity >98% minimizes false referrals (~1-2%) | Ordinal grading enables severity-based triage



Deployment Tradeoffs

Property	Two-Phase	Ensemble	Ensemble + TTA
Inference Passes	1	5	40
Storage	~48 MB	~240 MB	~240 MB
AUC-ROC	0.9992	0.9992	0.9994
Uncertainty	Not native	Pred. variance	Pred. variance
Best For	Mobile / Edge	Hospital	Research

Mobile / Edge

Two-Phase model: single pass, compact storage, ideal for resource-constrained settings

Hospital Systems

Ensemble: prediction variance provides uncertainty estimates for clinical decision support

Research

Ensemble + TTA: maximum accuracy with 40 passes for comprehensive evaluation



Limitations

Single Dataset

APTOS 2019 only — no external validation on IDRiD, Messidor-2, or EyePACS

Geographic Bias

Training data from rural India clinics — may not generalize to other populations

Class Imbalance

Grades 3-4 represent only ~13% of data — limited evaluation of severe cases

No Clinical Comparison

Not compared against ophthalmologists or commercial screening systems

Retrospective Design

No prospective clinical validation — real-world performance unverified

Acknowledging limitations demonstrates scientific rigor and strengthens the overall contribution



Future Work

External Validation

IDRiD, Messidor-2, EyePACS datasets

High

Prospective Trial

Clinical validation in real screening workflow

High

RETFound Backbone

Foundation model pretrained on retinal images

High

Mobile Deployment

ONNX/TFLite for edge inference

Medium

Multi-Task Learning

DR grade + lesion segmentation jointly

Medium

Federated Learning

Privacy-preserving training across hospital sites

Low

Summary of Contributions

1

Controlled Comparison

First head-to-head study: two-phase TL vs. K-fold ensemble on the same EfficientNet-B3 backbone and dataset — single model rivals ensemble

2

TTA + Threshold Synergy

Demonstrated additive, independent improvements from test-time augmentation and threshold optimization

3

Comprehensive Evaluation

configurations evaluated with AUC-ROC, bootstrap confidence intervals,

Thank You



Questions & Discussion

Uon Chamroeun

chamroeunuon127001@gmail.com

Royal University of Phnom Penh